

Vincent Louis Claes

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Ann Arbor, MI

EDUCATION

University of Michigan - Ann Arbor

Aug. 2022 – Aug. 2026

Cumulative GPA: 3.802/4.000

Bachelor of Science in Astronomy and Astrophysics | 3.950/4.000

Bachelor of Science in Interdisciplinary Physics | 3.856/4.000

Bachelor of Music in Cello Performance | 3.900/4.000

EXPERIENCE

Research Assistant

May 2024 – Present

University of Michigan Department of Astronomy

- Conducted photometric modeling with Prof. Monica Valluri and Dr. Behzad Tahmasebzadeh on ~ 100 simulations
- Developed a new method to estimate rotational kinematics from vertical structure (“X-shape”) of galaxies
- Achieved parameter estimations in minutes, improving efficiency by $100\times$ over previous fits
- Authored a manuscript in preparation for submission to a peer-reviewed journal

Research Assistant

Jan. 2024 – May 2024

University of Michigan Department of Astronomy

- Modeled velocity distributions of infalling mass in protostar simulations with Prof. Lee Hartmann
- Analyzed ~ 10 simulations to trace accretion flow and kinematic structures
- Developed a 3D adjustable visualization of protostellar environment for presentation and analysis
- One of three undergraduate flash-talk speakers in the *New Perspectives on Protoplanetary Disks* session, AAS 245

Data Management Student Fellow

Aug. 2024 – Present

University of Michigan Center for Academic Innovation (CAI)

- Analyzed 1+ million enrollments to develop an outreach-driven algorithm tracking course “health” and engagement
- Built and deployed 3 user-friendly online dashboards visualizing enrollment trends for institutional stakeholders
- Designed a data-merging GUI to unify 1,000+ course administrative and financial records across online platforms
- Authored an outreach report presented to 30+ CAI administrators on technology-driven learning impact

PUBLICATIONS

V. Claes, B. Tahmasebzadeh, M. Valluri et al. 2025, “Correlation Between the X-Shape Parameters and the Bar Parameters in Boxy/Peanut Barred Galaxies”, to be submitted to a peer-reviewed journal

PRESENTATIONS

2025 Jun. 9: 246th American Astronomical Society Meeting, Anchorage, AK | Chambliss Poster Competition Final Round | Research on edge-on boxy/peanut galaxy morphology and modeling

2025 Apr. 30: CAI Research Showcase, Ann Arbor, MI | Professional Showcase | Presentation on development and implementation of the “Course Health” analytics algorithm

2025 Mar. 27: Michigan Student Symposium for Interdisciplinary Statistical Sciences, Ann Arbor, MI | Interdisciplinary project integrating data science methods with astrophysical modeling

2025 Jan. 14: 245th American Astronomical Society Meeting, National Harbor, MD | One of three undergraduates invited for flash talk (1-slide), “New Perspectives on Protoplanetary Disks” Session

2025 Jan. 14: 245th American Astronomical Society Meeting, National Harbor, MD | Poster presentation on protostellar infall and early stellar evolution

HONORS AND AWARDS

- Chambliss Astronomy Achievement Student Award Finalist** Jun. 2025
- Finalist in the Chambliss Poster Competition at 246th American Astronomical Society Meeting
- University Honors (5x)** Dec. 2023 – Present
- Recognized for a cumulative GPA of 3.5+ across five consecutive semesters
- Rogel Award of Excellence** Aug. 2022 – Aug. 2026
- Recipient of a full-tuition merit scholarship awarded up to 75 U-M students annually

CERTIFICATIONS

- Angell Hall Planetarium & Observatory Operator** Nov. 2024
- University of Michigan Angell Hall*
- Conduct the *Zeiss ZKP 3/B projector* within 24-ft Astro-Tec planetarium dome for educational and public outreach sessions
 - Operate the *SkyQuest XT8 Dobsonian*, *Coronado Personal Solar Telescope*, and *0.4-m Ritchey-Chrétien reflector* for observational and instructional use

TECHNICAL SKILLS

Languages: Python, SQL, C++
Software & Systems: Tableau, LaTeX, Linux, Looker Studio, Git, MATLAB
Libraries: NumPy, SciPy, AstroPy, pandas, Matplotlib
Leadership: Principal Cellist of the University Philharmonia Orchestra (Aug. 2025 – Apr. 2026) — lead section in performances and rehearsals with faculty conductors and guest artists